Adversarial Patch Attacks and Defenses for Traffic-Light Classification

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INTRODUCTION

Motivation:

Misreading a red light as green can send a vehicle through an intersection. Detection is relatively stable, but the classifier that reads the light's state is vulnerable to physically realizable adversarial patches. Most adversarial research targets traffic signs or general object detection. Traffic-light recognition has received comparatively little attention.

Research Questions:

1. Can adversarial patches fool YOLO traffic-light classifiers?
2. Can adversarial training improve robustness?

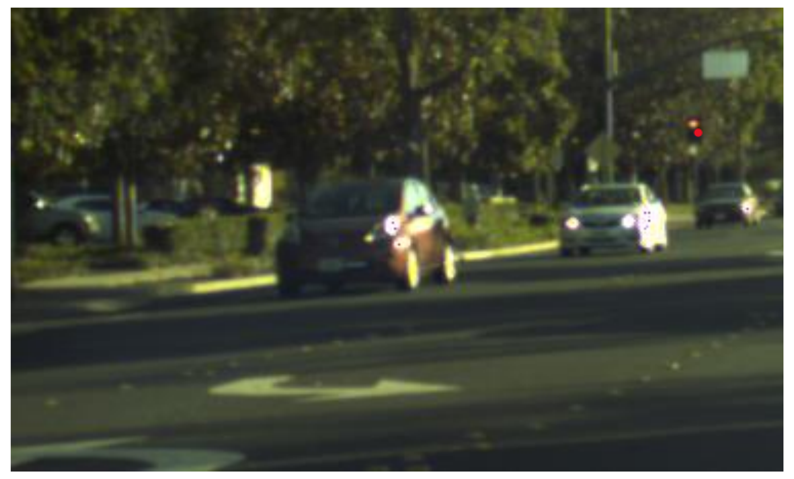

Figure 1. Clean BSTLD traffic-light image.

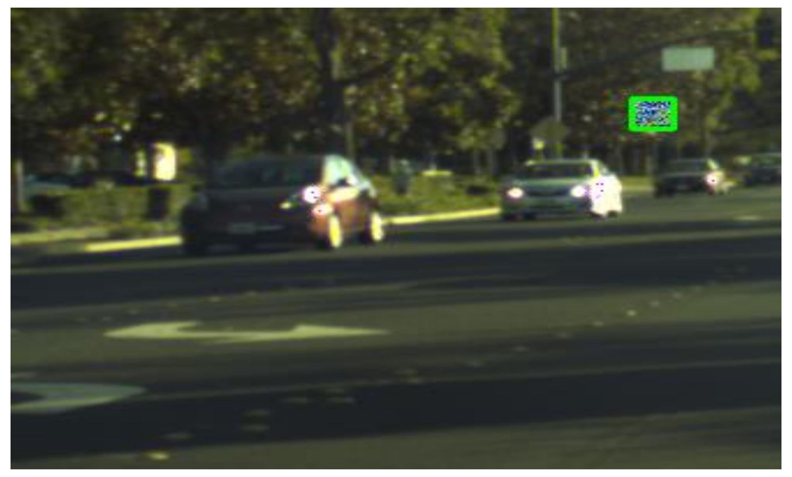

Figure 2. BSTLD image with EOT adversarial patch.

EXPERIMENTAL SETUP

Datasets	LISA (US) and BSTLD (EU): dashcam images with annotated traffic-light states
Model	YOLOv8m, COCO-pretrained, trained separately on each dataset
Threat Model	White-box (gradient access); patch is a printable sticker on the light
Adversarial Attack	Universal patch, targeted red → green. 125 × 125 px, scaled to the detected light. PGD (20 steps) under Expectation over **Transformation (EOT):** sampled over random rotation, brightness, translation, and scale so it survives real viewing conditions.
Defense	**Adversarial training:** frozen patch re-applied to training images with labels unchanged, so the patch must not alter the predicted state
Evaluation Metrics	**Classification Accuracy:** Correct light state, given the light was detected **Targeted Misclassification Rate (TMR):** red lights flipped to green by the attack **Vanishing Rate:** Lights not detected at all under attack

SYSTEM DESIGN

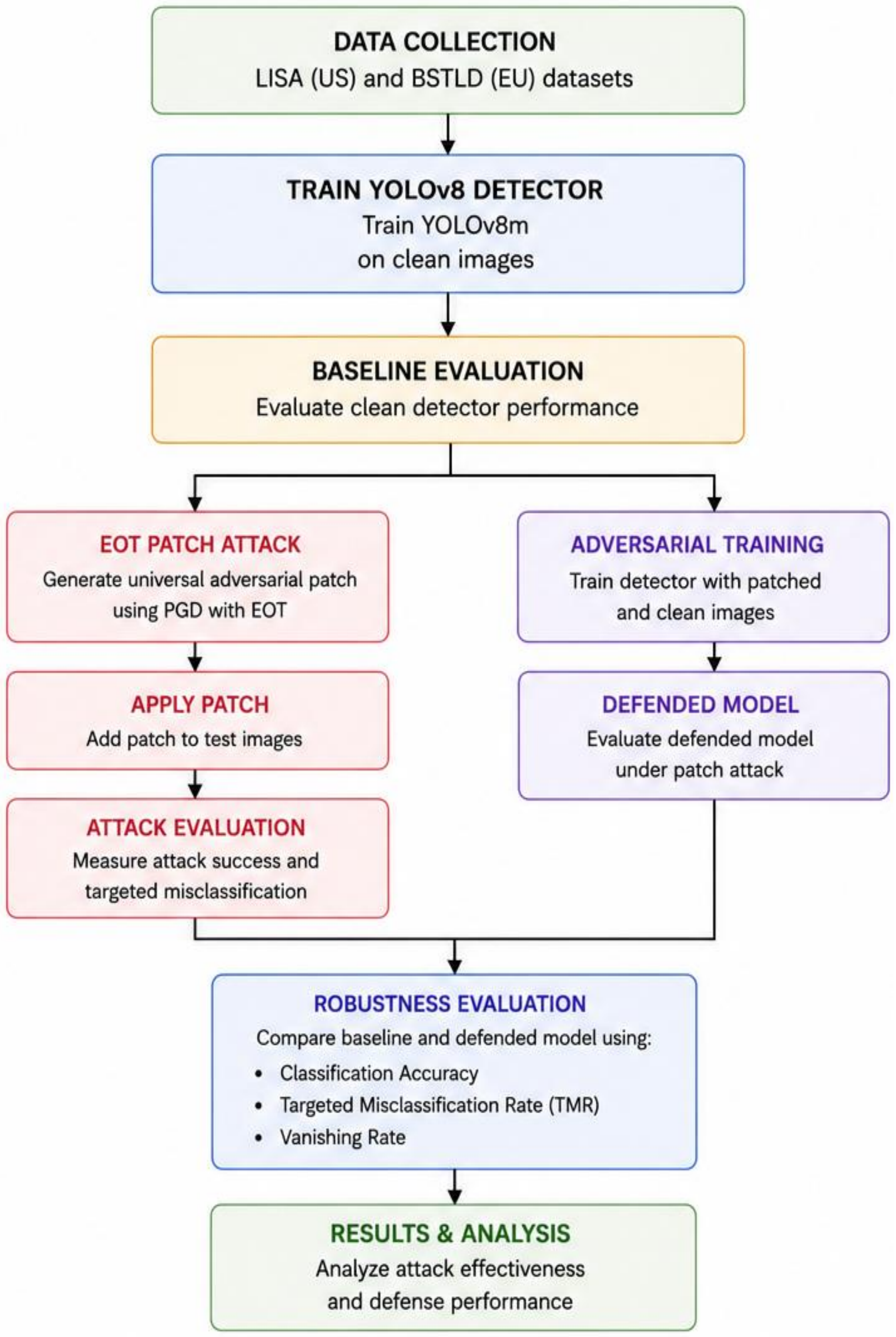

Figure 3. Adversarial attack, defense, and evaluation pipeline.

RESULTS

Dataset	Condition	Accuracy (%)	TMR (%)	Vanishing (%)
BSTLD	Clean	95.8	2.0	0.0
BSTLD	EOT Patch	23.9	72.9	3.2
BSTLD	Defense	76.9	19.6	3.5
LISA	Clean	96.4	2.0	0.0
LISA	EOT Patch	24.5	70.8	4.0
LISA	Defense	78.5	18.3	3.8

Figure 4. Performance metrics for BSTLD and LISA under clean, EOT patch attack, and adversarial defense conditions.

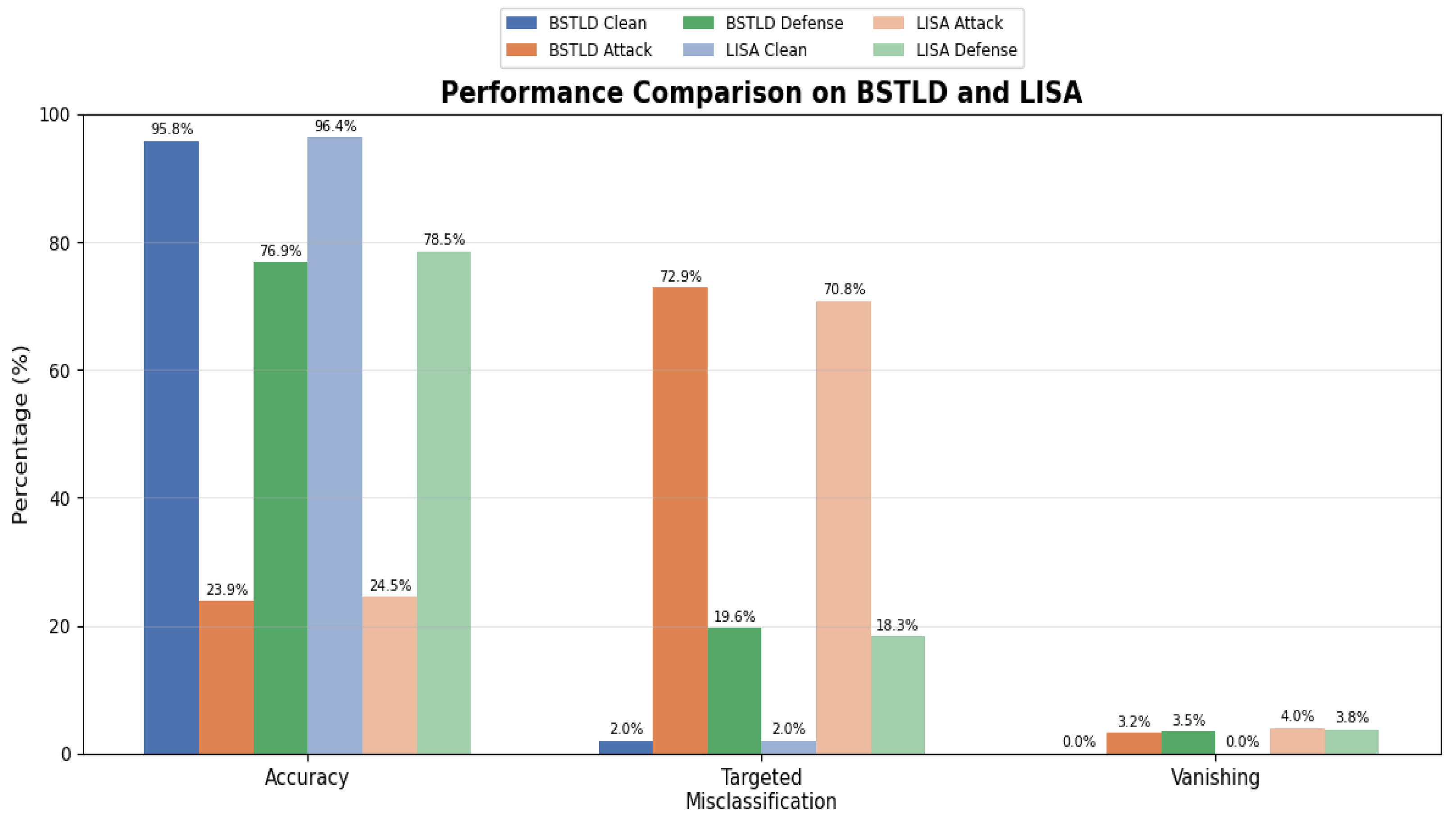

Figure 5. Performance under clean, EOT patch attack, and defense conditions.

- EOT patch reduced classification accuracy from **95.8%** to **23.9% (BSTLD)** and **96.4%** to **24.5% (LISA)**.
- Adversarial training reduced TMR from **72.9%** to **19.6% (BSTLD)** and from **70.8%** to **18.3% (LISA)**.

CONCLUSIONS

- EOT adversarial patches consistently induced targeted red-to-green traffic-light misclassification.
- Adversarial training improved robustness while preserving clean accuracy.
- Consistent trends across BSTLD and LISA indicate the approach generalizes across datasets.
- Future work includes adaptive attacks and real-world validation.

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